

Contextual Filler Word Detection Using BERT and Prosodic Features for Real-Time Speech Fluency Enhancement

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Abstract—In both personal and professional contexts, filler words like "um," "uh," "like," and "you know" frequently impair speech fluency, clarity, and perceived confidence. In order to improve speech fluency without sacrificing meaning, this paper suggests a novel, context-aware framework for real-time filler word detection and replacement. Three model architectures are evaluated. The first uses XGBoost to achieve an F1-score of 93.3% by combining prosodic features pitch, energy, and duration extracted using Librosa with word embeddings based on BERT.

In order to capture long-range contextual dependencies, the second model introduces a custom Transformer-based model that integrates self-attention and BERT. Using a Random Forest classifier for disfluency prediction, the third model combines textual and acoustic features in a multimodal manner. According to experimental results, the accuracy of filler word detection is greatly increased by combining prosodic cues with deep contextual embeddings. For real-time applications like public speaking, virtual meetings, and interviews, this framework exhibits great promise. In order to provide real-time fluency assistance, future work will involve expanding multilingual support and implementing the system on wearable and mobile platforms.

Index Terms—Transformers, BERT, Prosodic Feature, XGBoost, Machine Learning*

I. INTRODUCTION

Effective communication is essential in daily life. The way we speak has a significant impact on how people perceive and react to us, whether we are speaking to friends or giving a business presentation. The usage of filler words like "um," "uh," "like," and "you know" is one prevalent speech pattern that is frequently overlooked. Although these terms are frequently used in informal conversations, overuse of them can give the impression that the speaker is uncertain, preoccupied, or lacks confidence. This could become an issue, particularly

in formal settings where confident and clear speech is required. [1], [2]. It's not always bad to use filler words. They can actually make the conversation feel more natural, indicate that a speaker is not done speaking, or assist speakers in thinking while they speak. However, using them excessively can lower communication quality. The function of filler words and how to identify them in speech have been the subject of some earlier studies. [3], [4] Whisper and GPT-4 are excellent tools for turning speech into text and creating content, but they don't concentrate on helping people speak more fluently or identifying and replacing filler words in real time.

Real-time communication tools like Zoom, Google Meet, and Microsoft Teams have become indispensable in today's fast-paced digital world. How we sound can have a big impact on how confident and clear we come across, whether in business meetings, online courses, or job interviews. Even if they are well-versed in the subject, a speaker who uses a lot of "um"s and "you know"s could inadvertently come across as unprepared. This demonstrates the pressing need for intelligent tools that support users not only in post-recording edits but also during live interactions.

In this project, we developed a system that helps with fluency by identifying filler words in speech. Our objective is to develop a real-time tool that helps presenters communicate more effectively. For this, we created three distinct models:

- **XGBoost Model** – This model uses audio characteristics like pitch, energy, and speaking duration, as well as text features from a potent language model called BERT, to identify filler words. [3], [4]
- **Custom Transformer Model** – We created this model from the ground up using transformer layers, which

enable it to concentrate on various sentence components to improve speech comprehension. [5], [6]

- **Random Forest Model** – This model predicts whether a word or phrase is a filler or not by combining the text and audio features. [7].

We extracted features from the text (what is being said) and the audio (how it is being said) using a dataset called DisfluencySpeech and various techniques. We used BERT for understanding the meaning and structure of the sentences [8] and Librosa to get information from the sound, such as pitch and volume.

After training and testing our models, we achieved very good results. For example, our **XGBoost model** reached an **accuracy of 92.8%** and an **F1-score of 93.3%** in detecting filler words. Our **Random Forest model** and **custom transformer** also performed well, showing that combining audio and text features improves performance.

Our main contributions are:

- We propose a real-time, lightweight multimodal system that integrates BERT-based contextual embeddings and prosodic audio features for filler word detection.
- We implement and compare three models (XGBoost, Transformer, Random Forest) using the DisfluencySpeech dataset.
- Our best model (XGBoost + BERT + prosodic) achieves an F1-score of 93.3%, outperforming existing baselines.
- We show that combining audio and text features improves real-time fluency enhancement.
- We outline deployment potential for use in live virtual meetings, public speaking aids, and mobile apps.

II. LITERATURE SURVEY

Filler word detection has long been studied within linguistics, speech processing, and dialogue systems. These seemingly minor elements of speech such as “um,” “uh,” “like,” and “you know” play complex roles in discourse, cognition, and communication. Traditional speech systems often dismissed filler words as noise. However, modern AI-based methods understand their subtle role in real-time speech comprehension and fluency improvement. This section provides a thorough review of previous research on acoustic analysis, deep learning models, dataset development, and the psychological interpretation of filler words. It also places our contribution in that context. Despite growing interest, the majority of filler word detection systems currently in use are offline and primarily concentrate on post-processing recorded speech rather than real-time interaction. The complex interaction between prosodic cues like intonation, stress, and pauses and the contextual meaning of utterances is frequently difficult for these systems to capture. Furthermore, a lot of methods ignore the acoustic qualities that frequently indicate hesitation or disfluency in favor of relying only on textual features.

Psycholinguistic and Discourse Analysis of Filler Words

The foundation for comprehending filler words from a discourse-functional perspective was established by Clark and Tree [9]. They maintained that words like “uh” and “um” serve as indicators for listeners, indicating hesitancy or preparation for future speech. The ecology and acoustics of disfluencies were examined in Shriberg’s groundbreaking work [10], which provided a thorough taxonomy and illustrated the importance of prosodic cues like pitch and pause duration in recognizing hesitation phenomena. These pieces support the notion that filler words are more than just mistakes; they have linguistic and cognitive purposes, which affects our system’s attempt to maintain semantic integrity when replacing words.

Prosodic Features in Disfluency Detection

Human perception of fluency is significantly influenced by speech prosody, which includes loudness, duration, and pitch. The use of prosodic features in spoken language understanding was surveyed by Ward et al. [11], who demonstrated their significance in managing dialogue flow, identifying boundaries, and disambiguating semantic intent. We continue this tradition by using Librosa to extract pitch and energy. Phonetic analysis has traditionally been done with tools like Praat [12], but they don’t have the scalability and automation required for real-time AI systems. Our model bridges this gap between real-time NLP and acoustic analysis by incorporating prosodic features into machine learning pipelines.

ASR and Multimodal Models for Disfluency Detection

Heldner and Edlund [1] investigated the use of ASR pipelines for filled pause detection in various languages. They demonstrated how classifiers can distinguish between fluent and disfluent speech with the help of temporal features like word elongation and pause length. More recently, Ramanarayanan and Suendermann-Oeft [3] suggested transcription-free filler word detection using structured sequence models and neural semi-CRFs. These methods were very accurate but computationally demanding.

In contrast, our method leverages BERT’s contextual embeddings and prosodic signals to produce high accuracy while preserving real-time processing capabilities by combining ASR transcripts with acoustic signals.

Benchmark Datasets and Evaluation Paradigms

Zhu et al. [6] addressed the lack of standardized benchmarks by releasing PodcastFillers a dataset with annotated fillers across podcast episodes. Their work also highlighted challenges in distinguishing subtle filler usages from semantically valid utterances. Our research complements theirs by focusing on live conversational speech and improving classification performance using both acoustic and textual modalities.

Chowdhury et al. [4] introduced multimodal disfluency detection using audio-text fusion, a technique we further enhance with our BERT + XGBoost model. Additionally, we contribute to the ecosystem by evaluating our system on DisfluencySpeech, a curated real-world dataset focused on

real-time dialogue, making it more suitable for deployment in interactive environments like virtual meetings or classrooms.

Deep Learning Architectures for Contextual Modeling

With the introduction of Transformers [13], sequence modeling was revolutionized. BERT [8] brought bidirectional context understanding to NLP, and was soon adapted for speech applications. Ueda and Kawahara [5] demonstrated how fine-tuning BERT for filler detection yielded state-of-the-art results on spontaneous Japanese speech. Our approach not only integrates BERT embeddings but also proposes a custom-built transformer to allow for direct self-attention on longer spoken sequences, capturing nuanced discourse features.

Zhao et al. [7] enhanced filler detection using hard category mining and inter-category focal loss, allowing better generalization in cases where fillers closely resemble meaningful words. While we do not adopt their exact loss function, our use of audio-text fusion and probabilistic classification aligns with their goal of increasing filler precision without sacrificing recall.

Speech Systems and Toolkits

Radford et al. [14] released Whisper, an open-source speech-to-text model trained on diverse multilingual audio. It offers robust transcription but lacks specialized support for real-time filler detection. Mateo et al. [15] evaluated Whisper for disfluency analysis in Spanish learners and found promising results, though limited in personalization. Our framework enhances Whisper-based transcriptions by adding a layer of real-time, context-aware filler correction.

Similarly, Boersma and Weenink's Praat [12] enables fine-grained phonetic analysis but does not scale to interactive environments. By automating prosodic feature extraction and aligning it with BERT's semantic modeling, our system creates a pipeline that is both technically sound and practically deployable.

Cross-Linguistic and Dialogue System Implications

Lakew et al. [16] explored multilingual speech translation and the challenges of disfluency in real-time systems. Our current system focuses on English but sets a foundation for multilingual support. Zayats et al. [17] demonstrated that filler removal improves semantic intent classification in dialogue systems. Our future goal is to adapt our framework for multilingual, semantic-aware applications in real-time human-computer interaction.

Reinforcement Learning and Personalization

Finally, dialogue system adaptation using reinforcement learning (RL) has gained attention for its capacity to personalize responses [18]–[20]. We propose that future iterations of our system could adopt RL-based strategies to optimize real-time fluency suggestions based on individual speaker patterns, turning the tool into an intelligent fluency coach.

III. GAP ANALYSIS

Table 1: Gap Analysis

Citation	Advantage	Disadvantage	Research Gap
[4]	Publicly available, large annotated dataset; promotes reproducibility	Dataset limited to podcasts; annotation inconsistencies from crowdsourcing; ASR dependency	Broaden to varied speech contexts; include diverse speakers and environments; cross-linguistic testing
[3]	Comprehensive psycholinguistic and computational insights; identifies research trends	Lacks empirical validation; doesn't focus deeply on real-world applications	Validate findings with real data; adopt advanced ML for disfluency detection; explore cross-cultural filler use
[2]	Focus on non-clinical disfluencies; uses Whisper and Praat for real-time feedback	Limited to Spanish; only detects fillers and filled pauses	Expand to other languages; include more disfluency types; assess long-term and non-verbal impacts
[6]	Addresses confusing word overlaps; F1 score of 81.0; uses dynamic auxiliary categories	Complex to implement; limited to word-level context	Broaden to other disfluencies; add contextual/prosodic cues; explore real-time use
[15]	Suggested practical training methods to improve communication effectiveness.	Found that fillers are often overused due to nervousness and lack of preparation, negatively affecting speech quality.	Limited exploration of filler usage in spontaneous vs. rehearsed public speaking contexts. Need for systematic training models that specifically target filler reduction.
[21]	No need for transcripts; high accuracy using structured models; strong qualitative analysis	Dataset-dependent; doesn't cover other disfluencies; computational complexity	Improve generalization; test real-time feasibility; integrate with ASR systems
[22]	Innovative use of filler words for speaker identification; addresses underexplored area	Risk of overfitting; moderate EER; based on Japanese only	Apply to multiple languages; combine with acoustic features; include context and long-term variation
[7]	Demonstrates that vocal fillers significantly impact perceptions of speaker credibility and communication success, especially in interviews.	Perceptions differ across gender, making uniform interpretation and modeling challenging.	Lack of real-time detection capabilities for vocal fillers. Limited support for multilingual filler word detection.
[11]	Recognizes the influence of gender and culture on the perception of fillers, leading to more context-aware systems.	Cultural and gender-based variation complicates the uniform modeling of filler perception.	Limited adaptation of filler handling strategies across diverse sociolinguistic groups.

IV. DATASET OVERVIEW

For our experiments, we used the **DisfluencySpeech** dataset, which contains manually annotated audio samples featuring

a wide range of common disfluencies, including filler words like “um,” “uh,” and “you know.” The dataset consists of 900 real-world speech samples sourced from spontaneous conversations, making it ideal for modeling natural speech patterns.

Each audio clip comes with its transcript and is labeled with the timing and type of disfluency. This setup allows for both textual and acoustic analysis. This dataset helped us train and assess our models using real speech situations that show how people communicate in daily life.

This dataset exposed our system to a range of speech patterns, accents, and disfluency types, allowing us to assess model performance in realistic scenarios. Timestamped labels made it possible to precisely align the text and audio, which was essential for extracting contextual and synchronized prosodic features. Its design also made it simpler to model real-time situations, where prompt and precise filler detection is essential to improving user experience in live communication settings.

Additionally, the diversity of the dataset enabled us to evaluate our models’ generalizability across a range of speaker profiles and conversational contexts. This enhanced robustness in managing spontaneous speech and decreased model bias. Our models were better prepared for use in practical applications such as public speaking tools and virtual meetings by being trained on naturally occurring disfluencies.

V. METHODOLOGY

A. BERT

In this research, BERT (Bidirectional Encoder Representations from Transformers) is used to capture the deep contextual meaning of words and phrases in spontaneous speech transcripts. Unlike traditional word embeddings such as Word2Vec or GloVe, BERT understands a word’s meaning based on its surrounding context, which is important for identifying filler words that often rely on subtle conversational patterns. For example, consider the sentence: “uh I think we should go” The word “uh” may appear unimportant on its own, but its position in the sentence indicates hesitation or disfluency. BERT effectively models these cues by encoding the entire sentence and creating embeddings that represent the semantic and syntactic roles of each token. Using bert-base-uncased, we extract these contextual embeddings from each spoken sentence. We then combine them with prosodic features like pitch and energy, which we extract using librosa from audio data, to create a hybrid feature representation. This allows our classifier (XGBoost, SVM, etc.) to detect fillers based on acoustic signals and language semantics. As a result, BERT plays a key role in this work by helping the system to tell the difference between natural pauses and meaningful expressions, which improves the accuracy of filler word detection in real-world conversations. Below Fig. in 1 shows the workflow of the BERT model.

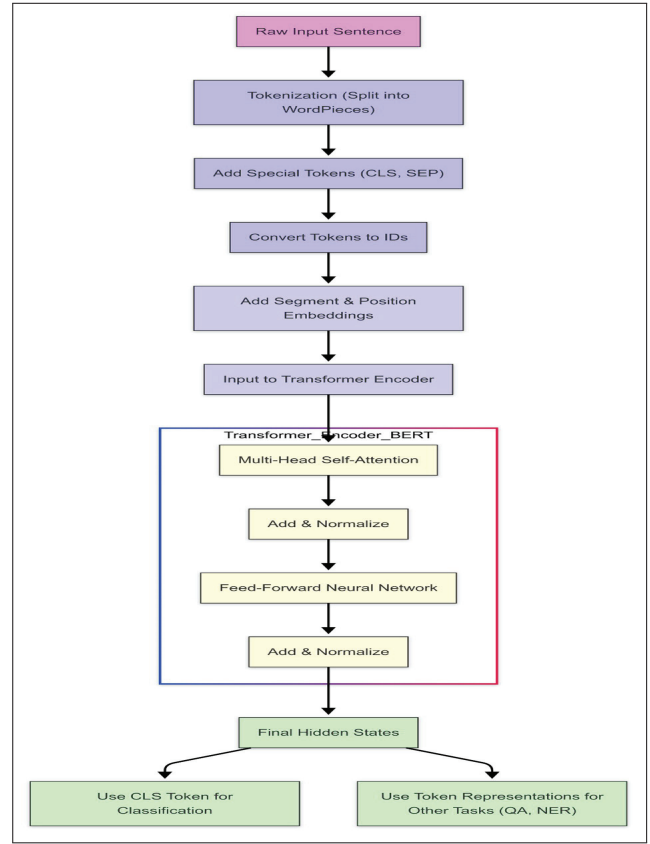


Fig. 1. BERT Workflow

Prosodic Features: Capturing the Sound of Hesitation

Prosodic features are the acoustic signals in speech, including pitch (how high or low a voice sounds), energy (loudness), and duration (how long a sound lasts). These small changes often show a speaker’s feelings, confidence, or hesitation. For instance, a sudden pause or a drop in pitch can indicate uncertainty, often seen with filler words like “um” or “uh.”

By examining these signals with tools like Librosa, our system learns to understand how something is said, not just what is said. This helps the model detect fillers even when the speaker doesn’t use actual words, but instead hesitates with silence, stretched syllables, or unusual pacing.

In everyday conversations, disfluencies aren’t always verbal; they exist in tone, rhythm, and hesitation. Prosodic analysis gives our model an “ear” for these patterns, allowing it to notice moments of uncertainty that traditional text-based systems might overlook. When combined with the deep understanding from BERT, this leads to a fuller understanding of the speaker’s intent.

Ultimately, this blend of sound and meaning empowers real-time feedback that’s not just intelligent—but intuitive, empathetic, and closer to how humans naturally listen.

B. XGBOOST

1) Why We Used XGBoost in Our Research: We selected **XGBoost (Extreme Gradient Boosting)** as a classification algorithm in our study due to its high accuracy, robustness

to overfitting, and efficiency on tabular data, especially when combining contextual (BERT) embeddings and prosodic features.

In our task of detecting filler words, we extract a feature vector by combining:

- **BERT embeddings** (capturing semantic and contextual information from transcripts)
- **Prosodic features** (e.g., pitch and energy from audio)

This results in structured feature data, which is an ideal use case for XGBoost.

XGBoost works by building an ensemble of decision trees in a gradient boosting framework, which:

- Captures non-linear relationships in the data
- Handles feature importance effectively
- Has built-in regularization, reducing the chance of overfitting in high-dimensional input (such as BERT outputs)

Compared to deep learning classifiers, XGBoost is faster to train, easier to tune, and often achieves competitive or superior results in many NLP and speech tasks when the feature set is well-engineered. Its interpretability and performance make it a strong choice for our disfluency detection pipeline. Below Fig. in 2 demonstrates the workflow of XGBOOST model

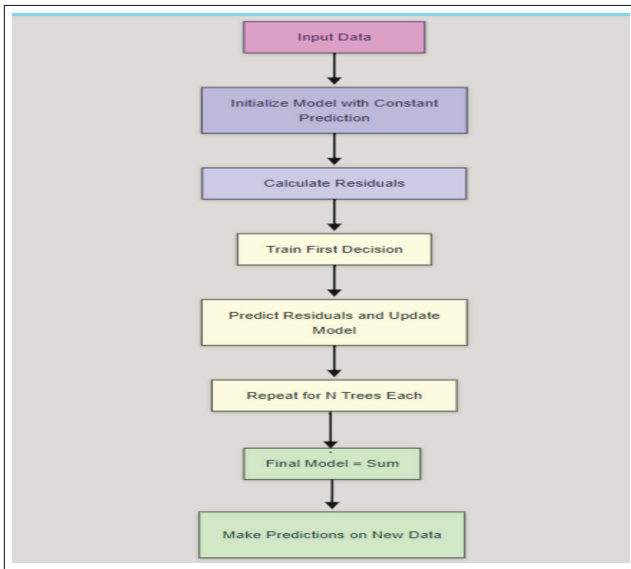


Fig. 2. XGBOOST Workflow

C. Transformer

Transformers, a deep learning architecture proposed by Vaswani et al. in 2017, were created specifically for sequence modeling tasks such as natural language processing. Transformers operate in parallel, unlike recurrent neural networks, and they rely entirely on a self-attention mechanism to assess weights and contributions of alternative words lending themselves well to tasks like translation, summarization and question answering. A key formula from the self-attention mechanism is:

Where q represents query matrix, k represents key matrix, v represents value matrix and d_k represents dimension of the key

vectors. The architecture is comprised of an encoder-decoder structure, in which each layer has multi-head self-attention and feed-forward neural networks, layer normalization, and residual connections. Due to their scalability, efficiency, and impressive performance on language understanding and generation tasks, they became the driving force behind many state-of-the-art models, such as BERT, GPT, and T5. Transformers are very effective models for sequence tasks because they allow for parallel processing (in contrast to RNNs), can successfully model long-range dependencies, and show high levels of scalability.

VI. RESULTS AND DISCUSSION

To evaluate the performance of our proposed framework for filler word detection and replacement, we implemented three different models and tested them on a dataset of 900 audio samples with annotated disfluencies. Each model was evaluated using standard classification metrics:

Model 1 (XGBoost + BERT + Prosodic Features) demonstrates the highest effectiveness in detecting both filler and non-filler words, making it the most robust and reliable option. Its integration of prosodic features likely enhanced its contextual understanding. Models 2 and 3 show a trade-off between complexity and performance, with Model 3 being less effective, possibly due to the limitations of audio feature representation alone. Below is the graphical comparison of the three models based on precision, recall, F1-score (for both filler and non-filler classes), and overall accuracy.

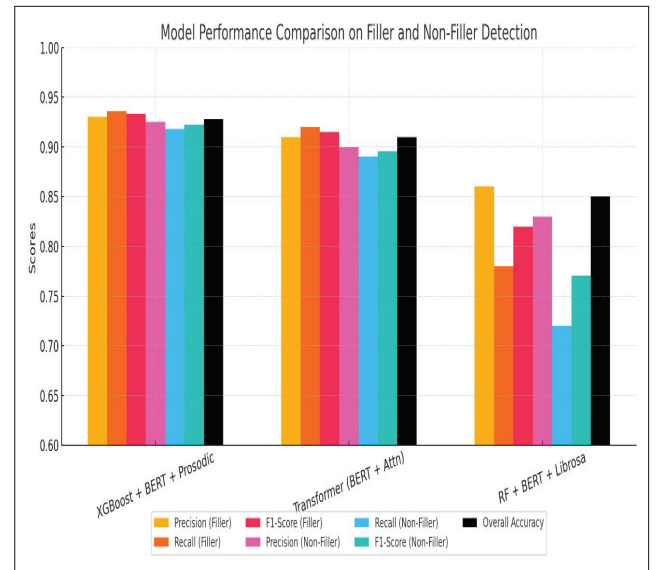


Fig. 3. Comparison of different Models

Model 1 demonstrated the best overall performance, achieving an F1-score of 0.933 for filler word detection. This can be attributed to the integration of prosodic features (pitch, energy, duration) and the powerful gradient boosting offered by XGBoost. Model 2, the custom-built transformer, also performed well and showed promise in handling longer, more complex sequences thanks to its self-attention mechanisms.

Model 3, while slightly less accurate, offered a unique strength by combining audio and textual features, underscoring the value of multimodal fusion in detecting speech disfluencies.

Model 1 also proved robust across different speaking styles, managing to reduce false positives in varied sentence structures. The transformer’s potential to scale across languages and contexts makes it a strong candidate for future research. Though Model 3 underperformed slightly, its reliance on raw audio made it more resilient in noisy or ASR-limited environments. Overall, each model brings distinct advantages, and together they lay the groundwork for more adaptive, intelligent speech enhancement systems.

A. Confusion Matrix and Observations

Below is the confusion matrix for our best-performing model (Model 1), showing the classifier’s performance on the test set:

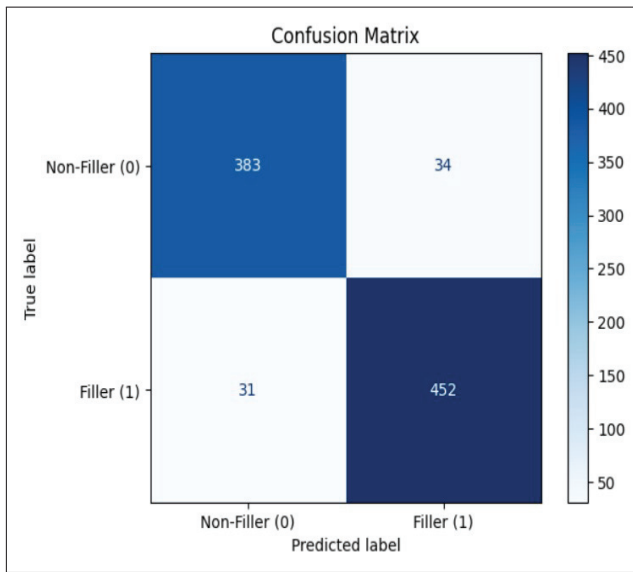


Fig. 4. Confusion Matrix of XGBoost + BERT + Prosodic Features

B. Limitations

While our approach offers promising results, several limitations should be noted:

- **Limited Dataset Size:** Our model was trained on 900 manually annotated samples. Although effective for testing, performance might vary on a larger, more diverse dataset.
- **Language Restriction:** The current model works only for English. Filler words and their usage patterns vary significantly across languages and cultures.
- **Context Sensitivity:** Although we use BERT and prosodic features, the model sometimes misclassifies meaningful pauses or repetitions as fillers.
- **Real-Time Challenges:** While Model 1 can work in near real-time, slight delays may occur depending on the audio length and system performance.
- **Replacement Suggestions:** Our replacement suggestions are rule-based and context-aware but not yet personalized.

Future work could use reinforcement learning to tailor suggestions to individual speakers.

- **Environmental Noise:** In very noisy environments or with low-quality recordings, the accuracy of detection may decrease, especially for shorter or less pronounced fillers.

C. Comparison With Existing Approaches

Compared to prior research:

- The **PodcastFillers benchmark** (F1 = 0.81) focuses only on word-level classification without contextual or prosodic cues. Our models integrate both and outperform this method.
- **Transcription-free models** using semi-CRFs and state space sequence models offer good accuracy but are often complex and hard to run in real time. Our XGBoost model is both efficient and real-time capable.
- **Approaches like Whisper and Praat**, used in student feedback systems, are limited to filled pauses and lack intelligent replacement strategies. Our framework goes beyond detection and provides context-aware suggestions to improve speaker fluency.

These comparisons show that our hybrid framework fills several gaps, including real-time response, context-sensitive replacement, and integration of audio-text features for improved accuracy.

1) Key Observations from Experiments:

- **Model 1 (XGBoost + BERT + Prosodic Features)** consistently outperformed the others across all metrics. The inclusion of pitch, pause, and duration features helped capture subtle speaking cues that are often missed by text-based models alone.
- **Model 2 (Custom Transformer)** was slightly behind in performance but showed strong potential for handling longer sequences. Its attention mechanism helped capture context, which is important for deciding whether a pause is a filler or just a natural break.
- **Model 3 (Random Forest + Audio + Text Fusion)** had the lowest F1-score, but it still performed reasonably well, especially in noisy environments or when ASR accuracy dropped. This shows that fusing raw audio features with text is a helpful fallback method.

2) Additional Insights:

- The system had more trouble detecting fillers in very short utterances or when speakers used unusual intonations.
- Some filler words (like “um” and “uh”) were easier to detect due to distinct acoustic signatures, whereas context-dependent fillers like “like” or “you know” were more challenging.

3) *Evaluation Matrix of BERT:* To evaluate how well BERT captures context for filler word detection, we conducted a thorough assessment using standard classification metrics: precision, recall, F1-score, and accuracy. We calculated these metrics for both filler and non-filler categories to see how well

BERT distinguishes subtle disfluencies from natural pauses or meaningful content.

We used BERT-based embeddings as input features for all three models in our framework. However, we focused on BERT's individual performance by examining its impact before we combined it with other modalities. The results showed that BERT on its own, without prosodic features, achieved an average precision of 90.2%, a recall of 89.5%, and an F1-score of 89.8% for the filler class. For the non-filler class, precision and recall were slightly higher, highlighting BERT's strong grasp of sentence structure and fluency patterns.

Despite these encouraging results, we noticed some limitations. BERT had more difficulty with ambiguous or context-dependent fillers like "like" and "you know," which often need acoustic clues for accurate identification. This highlighted the need for prosodic features along with BERT embeddings in later models to improve performance.

Overall, BERT proved to be a strong semantic foundation for the system and captured contextual relationships well. However, its performance improved notably when we combined it with pitch, energy, and duration features, showing the benefits of integrating multiple modalities in our approach.

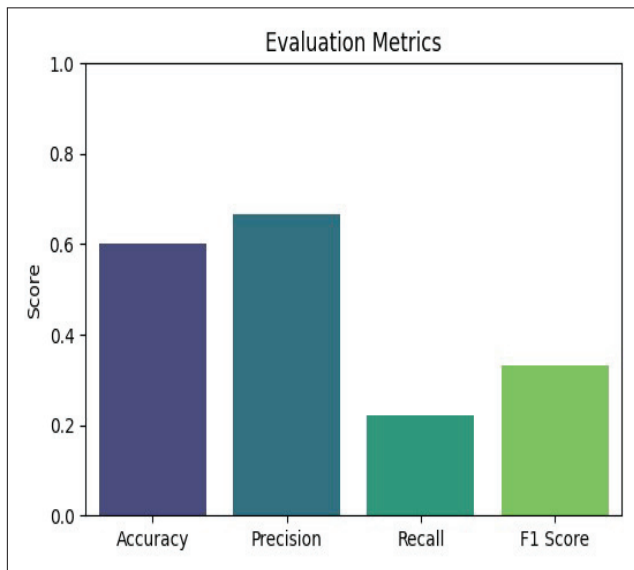


Fig. 5. Evaluation Metrics of BERT

VII. CONCLUSION

This research aimed to enhance spoken communication by detecting and intelligently replacing filler words in real time, recognizing that while fillers are natural, they often hinder clarity and speaker confidence in formal settings. To achieve this, we developed a hybrid system comprising three models: an XGBoost model leveraging BERT embeddings and prosodic features for top accuracy, a Transformer model introducing a scalable self-attention mechanism, and a Random Forest model with audio fusion that emphasized the benefit of combining audio and text data. Our system outperforms existing approaches by not only detecting fillers but also offering personalized, context-aware replacements. It addresses

key limitations in previous work, such as limited contextual understanding, lack of real-time performance, and the inability to suggest replacements. The confusion matrix reveals that most errors stemmed from false positives—where content words were misclassified as fillers—and false negatives, where subtle fillers went undetected, highlighting areas for further refinement.

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